



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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<04/09/2026>



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies;
 - Data collection using SpaceX API.
 - Data collection using web scraping.
 - Data Wrangling.
 - Exploratory data analysis using SQL.
 - Exploratory data analysis through visualizations.
 - Interactive visual analysis using Folium.
 - Interactive dashboard creation with Plotly Dash.
 - Building and deployment of machine learning prediction models.
- Summary of all results;
 - Success rate is observed to generally improve over time.
 - The launch site KSC LC-39A is found to have the highest mission success rate.
 - The orbit types ES-L1, HEO, SSO and GEO are discovered to have the most successful mission outcomes.
 - In the earlier years, heavier payloads have a high failure rate but they improve over time.
 - The DecisionTreeClassifier algorithm outperforms the rest of the models by achieving the highest accuracy in predicting landing outcomes.

Introduction

- In today's age, SpaceX offers the best cost for launching rockets, with the Falcon 9 being priced at around 62 million dollars.
- This is far cheaper than their closest competitor who charge upwards of 150 million dollars.
- SpaceX is able to achieve such reduced costs due to its capability to land and reuse their rockets.
- Given this scenario, if we are able to predict whether the rocket will land, we can therefore gain insight on potential costs. This will allow involved parties to make an informed decision whether to bid or not to bid on a SpaceX rocket.
- Therefore, the main aim of this project is to make predictions on the successful landing of SpaceX Falcon 9 rocket, which is a vital aspect in determining the cost and efficiency of reusable rockets.

Section 1

Methodology

Methodology

Executive Summary

- **Data collection methodology:**
 - Extraction of rocket launch data from SpaceX API using GET requests.
 - Extraction of tabular SpaceX launch data from Wikipedia using web scraping.
- **Perform data wrangling:**
 - Filtering and extraction of relevant records.
 - Handling missing values, flattening fields and assigning labels.
- **Perform exploratory data analysis (EDA) using visualization and SQL:**
 - Applying libraries such as Pandas and Matplotlib in visualizing the relationship between variables.
 - Using SQL queries to assess and manipulate the data.

Methodology

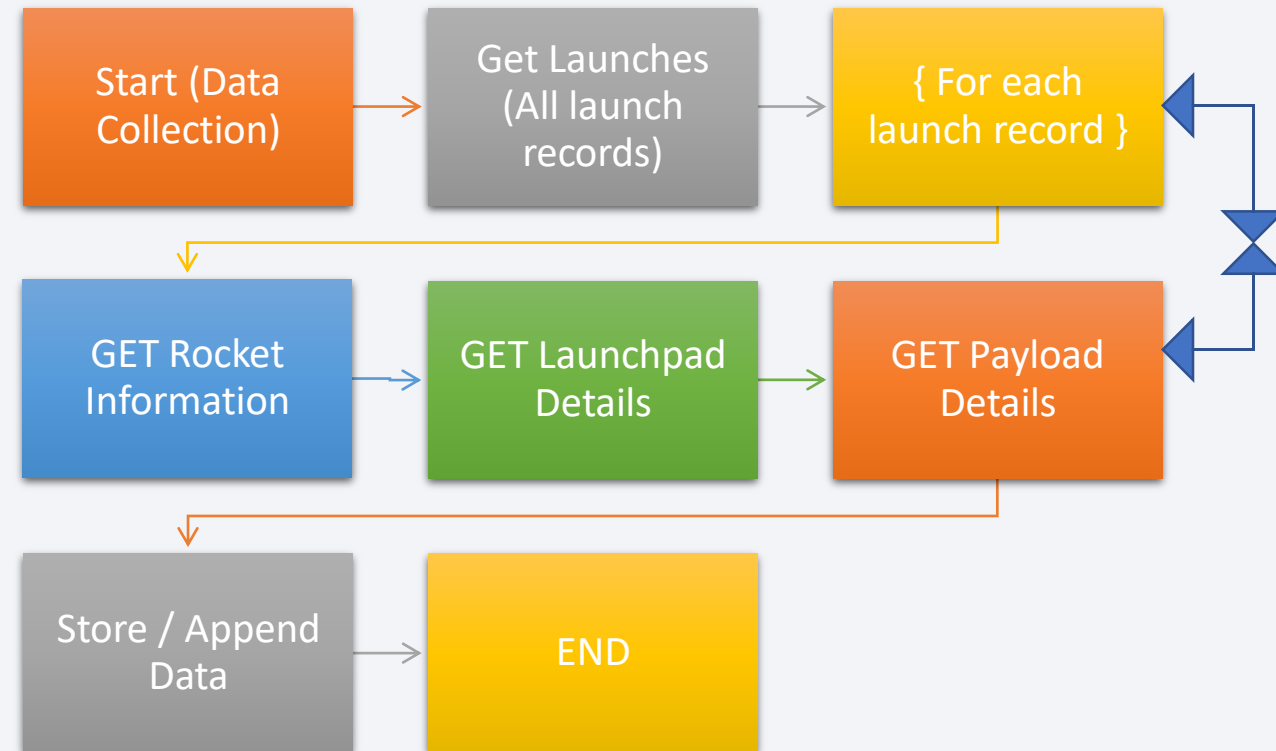
- Perform interactive visual analytics using Folium and Plotly Dash:
 - Performing geospatial analytics using Folium.
 - Building an interactive dashboard using Plotly Dash.
- Perform predictive analysis using classification models:
 - Applying the Scikit-Learn package to pre-process and split the data, train different models and tune hyperparameters.
 - Evaluating the models using metrics accuracy and confusion matrices.

Data Collection

- Launch data is retrieved from the SpaceX API using GET requests method and includes information such as;
 - Rocket used.
 - Payload delivered.
 - Launch specification.
 - Landing specification.
 - Landing outcome.
- Historical launch records of Falcon 9 rocket are collected from Wikipedia using web scraping.

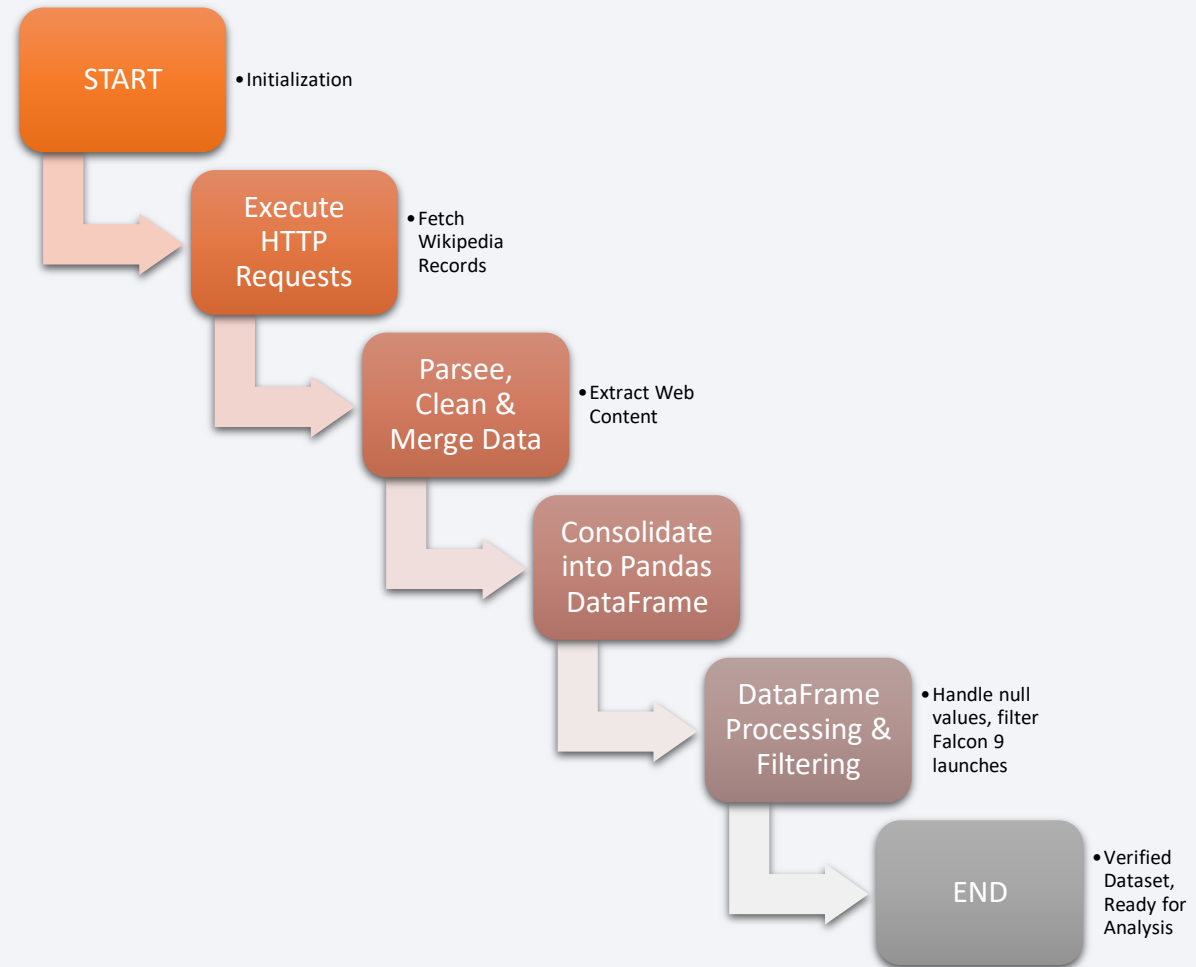
Data Collection – SpaceX API

- This process involves:
 - Making a GET request to the SpaceX REST API.
 - Converting the response to JSON.
 - Converting the JSON file into a Pandas dataframe.
- https://github.com/MbathiTwain/IBM_DATA-SCIENCE-CAPSTONE-PROJECT/blob/main/01-data-collection-api.ipynb



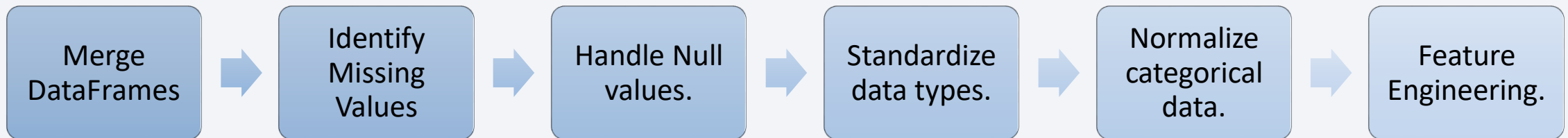
Data Collection - Scraping

- This process involves:
 - Retrieving the HTML source of Falcon 9 launch page from Wikipedia using HTTP Get.
 - Extracting the tables using a BeautifulSoup object.
 - Populating the Pandas DataFrame using the extracted data.
- https://github.com/MbathiTwain/IBM_DATA-SCIENCE-CAPSTONE-PROJECT/blob/main/jupyter-labs-webscraping.ipynb



Data Wrangling

- During this process, the following steps are carried out;
 - Identifying missing values in each variable as a percentage.
 - Revealing whether columns are numerical or categorical.
 - Discovering the number of launches per site, as well as the distribution of orbit type.
 - Grouping the landing outcomes by binary outcomes and storing them as a class to be used as the target value.
- https://github.com/MbathiTwain/IBM_DATA-SCIENCE-CAPSTONE-PROJECT/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb



EDA with Data Visualization

- This step involves creating a number of plots and charts that better describe the data;
 1. Scatter charts: This plot is used to visualize the relationship between different numeric features in our data. It can also reveal if the variables are correlated.
 2. Bar charts: This chart is used to compare numerical and categorical features in the data. In our case, it is applied to show the relationship between Success rate and Orbit type.
 3. Line charts: This chart can be used to show the changes in our features over time. In our case, we plot a line chart to show the change in success rate across the years.

https://github.com/MbathiTwain/IBM_DATA-SCIENCE-CAPSTONE-PROJECT/blob/main/edadataviz.ipynb

EDA with SQL

- In order to further understand our data, we perform a number of SQL queries that;
 - Display the unique launch sites.
 - Show 5 records of launch sites beginning with 'CCA'.
 - Show total payload mass carried by boosters that NASA (CRS) launched.
 - Display the average payload mass carried by booster version F9 v1.1.
 - Show the data when the first successful landing on a ground pad occurred.
 - List the booster versions with successful landings on the drone ship having payload mass between 4000Kg and 6000Kg.
 - Show the number of failed and successful outcomes.

EDA with SQL

- Display the booster versions that carried the maximum payload mass.
- Show the failed landings on drone ships with their booster versions, and launch site names for the year 2015.
- Display the distribution of landing outcomes between June 4th, 2010 and March 20th, 2017. Arranging them in descending order.
- https://github.com/MbathiTwain/IBM_DATA-SCIENCE-CAPSTONE-PROJECT/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

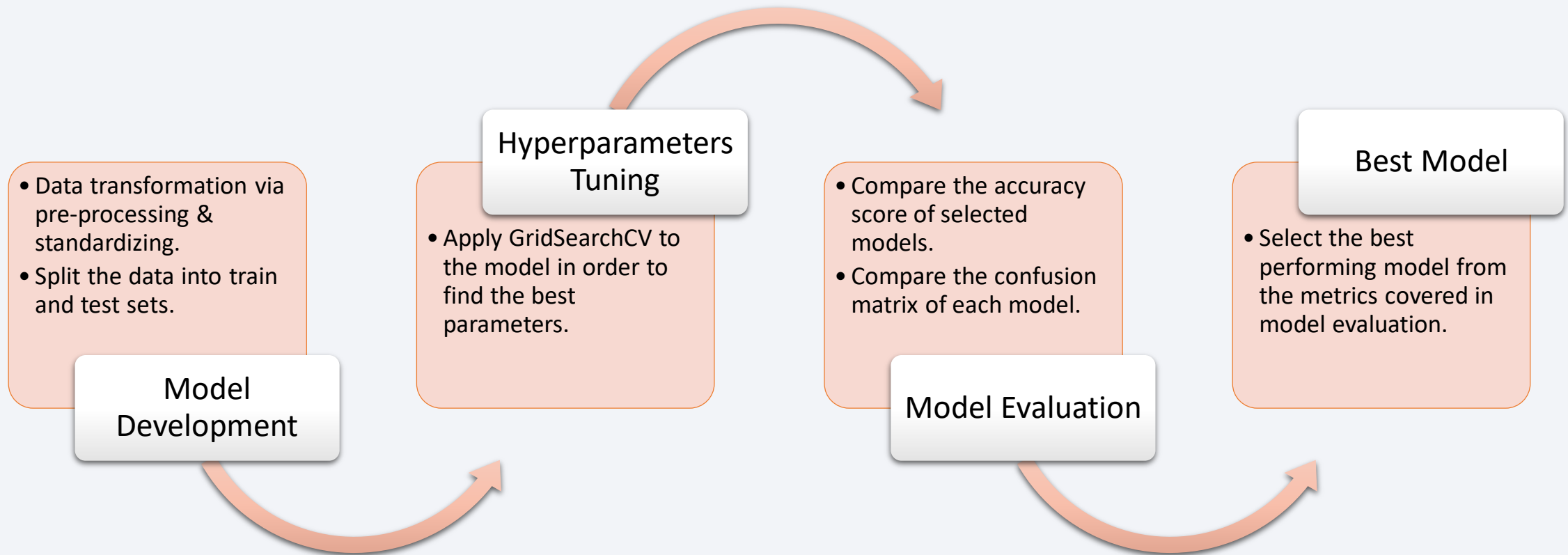
Build an Interactive Map with Folium

- In order to reveal geographical patterns in our data, a number of markers have been created on a map that display;
 1. Launch sites: Each launch site can be seen where the folium markers have been placed. Folium circles have also been added for better clarity.
 2. Successful and failed launches: Launches on the same launch site have been clustered together with successful and failed launches differentiated using different colored markers.
 3. Distance between launch sites and its proximities: A distance line is created that reveals the distance between a launch site and various landmarks such as railway line.
- [https://github.com/MbathiTwain/IBM_DATA-SCIENCE-CAPSTONE-PROJECT/blob/main/lab_jupyter_launch_site_location%20\(1\).ipynb](https://github.com/MbathiTwain/IBM_DATA-SCIENCE-CAPSTONE-PROJECT/blob/main/lab_jupyter_launch_site_location%20(1).ipynb)

Build a Dashboard with Plotly Dash

- A Plotly dashboard has been developed to facilitate interactive data exploration which includes;
 - A dropdown selector for choosing a launch site with an option of selecting all.
 - A pie chart which displays the breakdown of successful outcomes for selected sites.
 - A scatter plot which displays the landing outcomes by payload mass and booster version.
 - A payload mass range selector for filtering points on the scatter plot.
- [https://github.com/MbathiTwain/IBM_DATA-SCIENCE-CAPSTONE-PROJECT/blob/main/spacex-dash-app%20\(1\).py](https://github.com/MbathiTwain/IBM_DATA-SCIENCE-CAPSTONE-PROJECT/blob/main/spacex-dash-app%20(1).py)

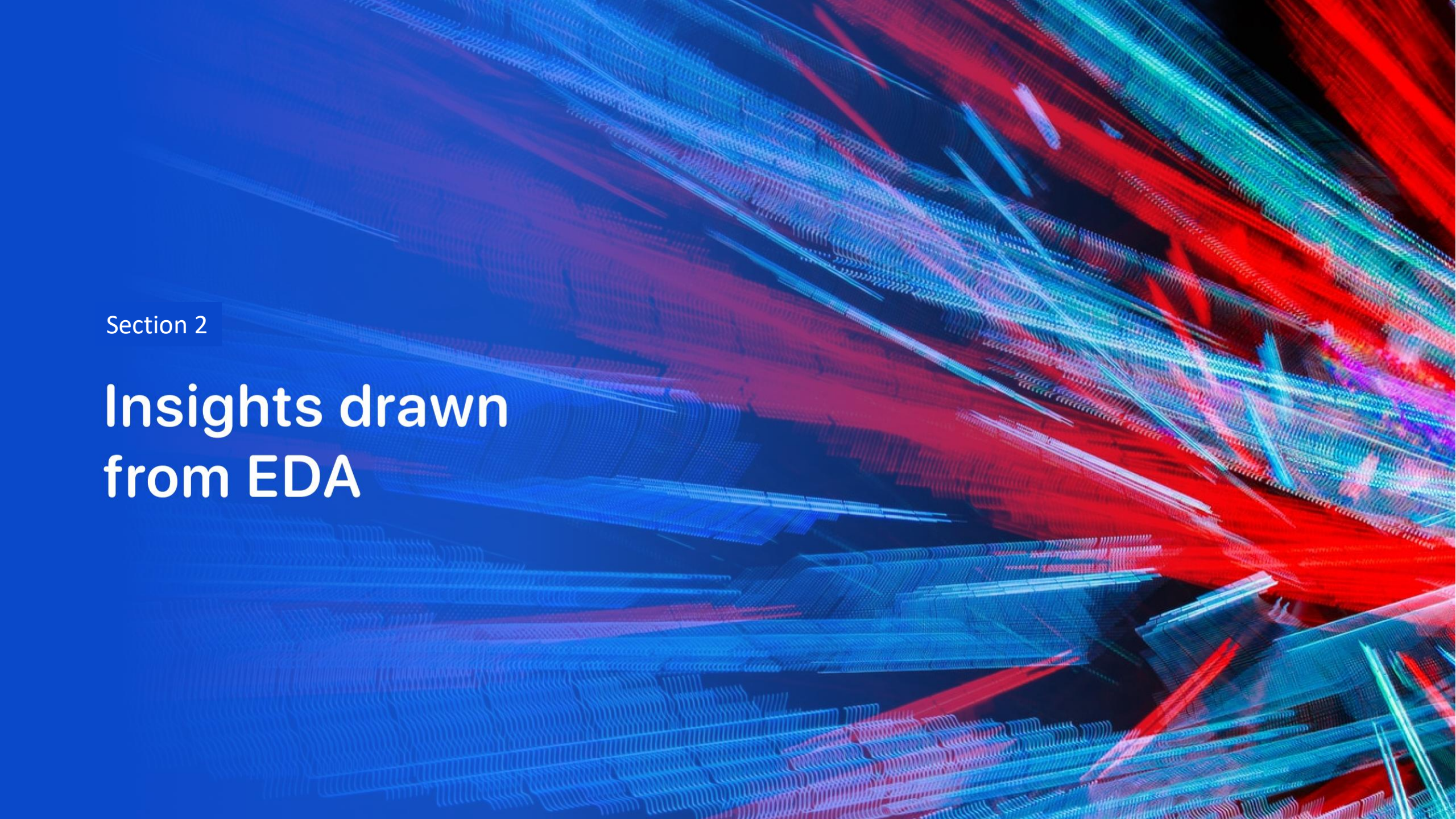
Predictive Analysis (Classification)



- [https://github.com/MbathiTwain/IBM_DATA-SCIENCE-CAPSTONE-PROJECT/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20\(1\).ipynb](https://github.com/MbathiTwain/IBM_DATA-SCIENCE-CAPSTONE-PROJECT/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20(1).ipynb)

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

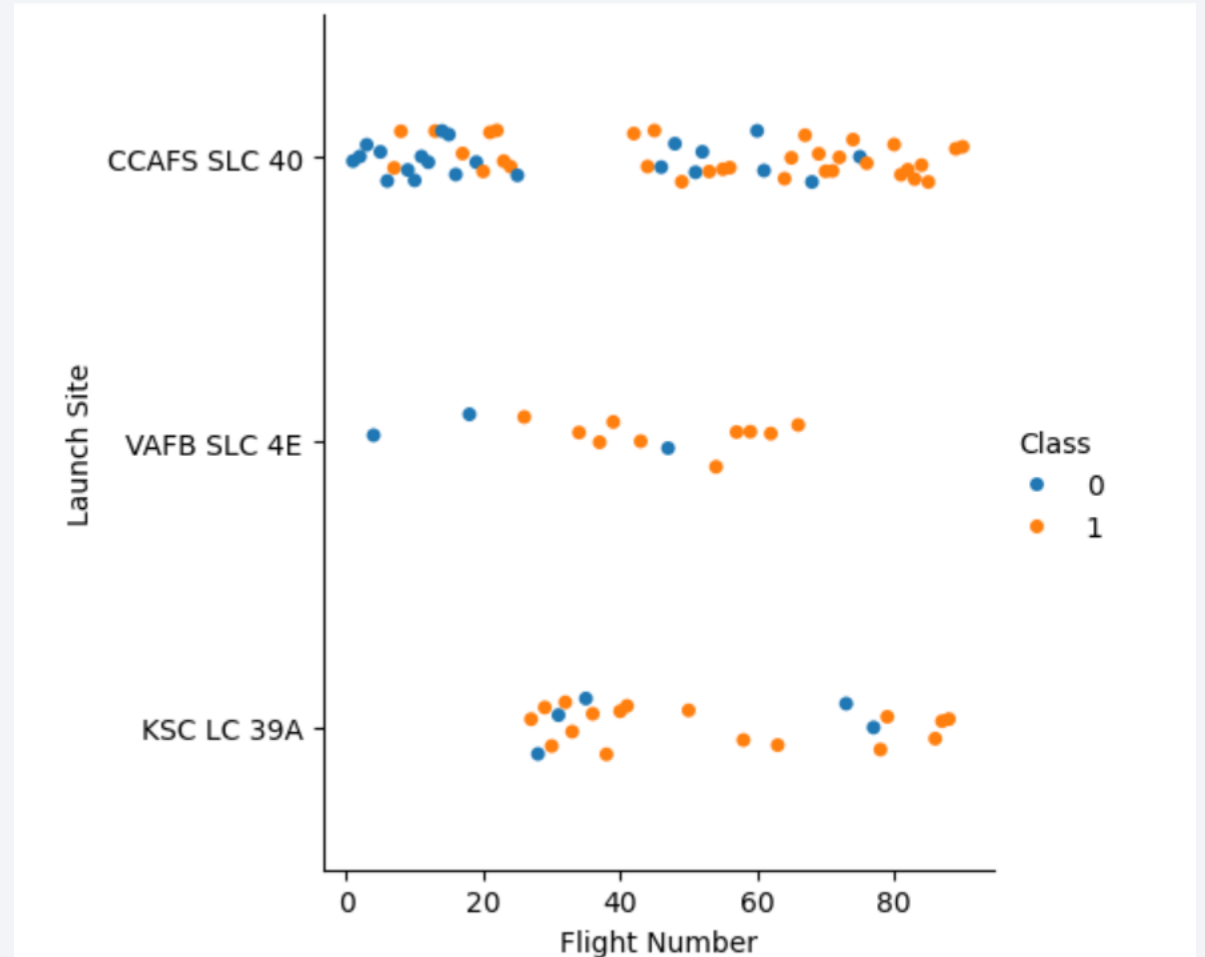
The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

Section 2

Insights drawn from EDA

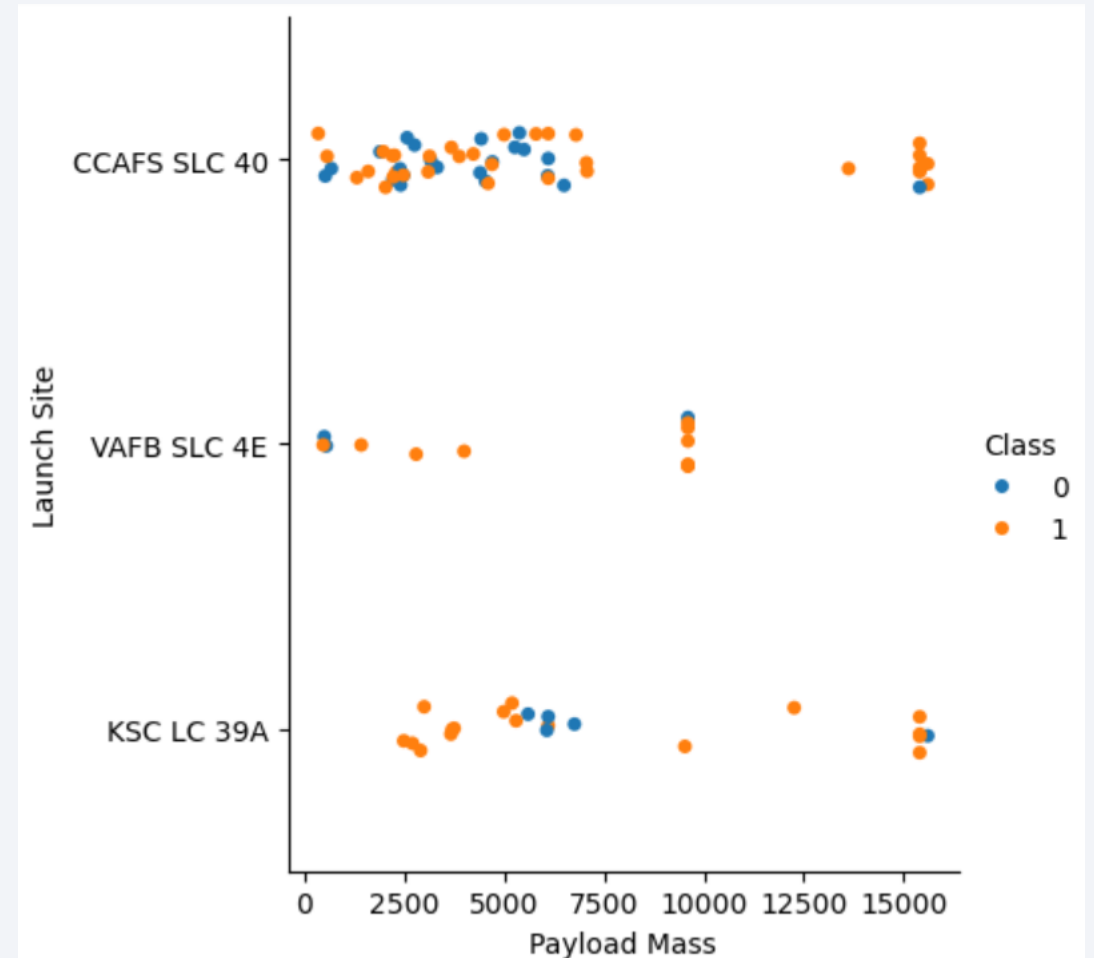
Flight Number vs. Launch Site

- This figure shows a scatter plot of Flight Number vs. Launch Site.
- it reveals that a high number of earlier flights (flight number < 30) were unsuccessful.
- It also reveals a general increase in success rate with increase in flight number.
- Launch site VAFB SLC 4E has the highest proportion of successful landings.



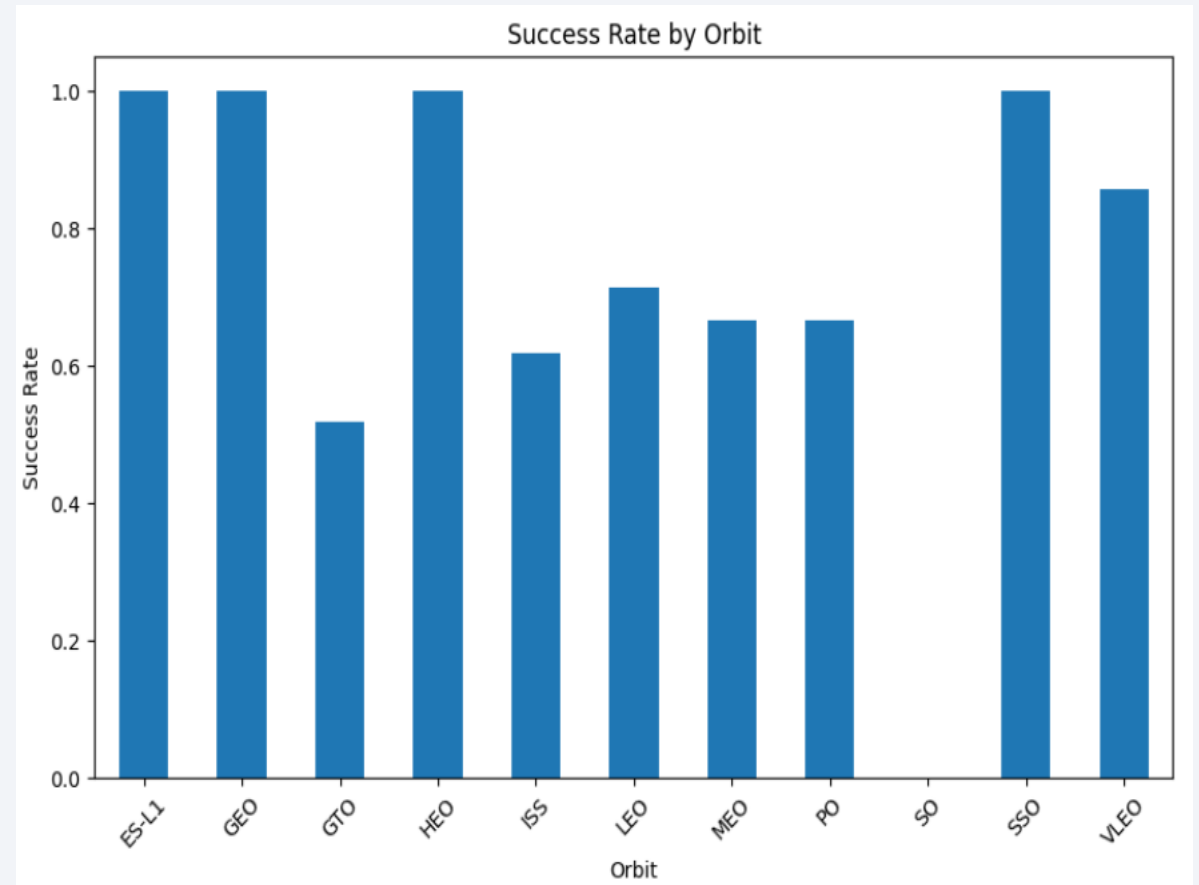
Payload vs. Launch Site

- This figure shows a scatter plot of Payload vs. Launch Site.
- It reveals that instances with a heavier payload mass have a higher success rate.
- Launch site VAFB SLC 4E handled rockets with a lighter payload compared to the other two.
- It also reveals that there is no significant correlation between payload mass and success rate given launch site.



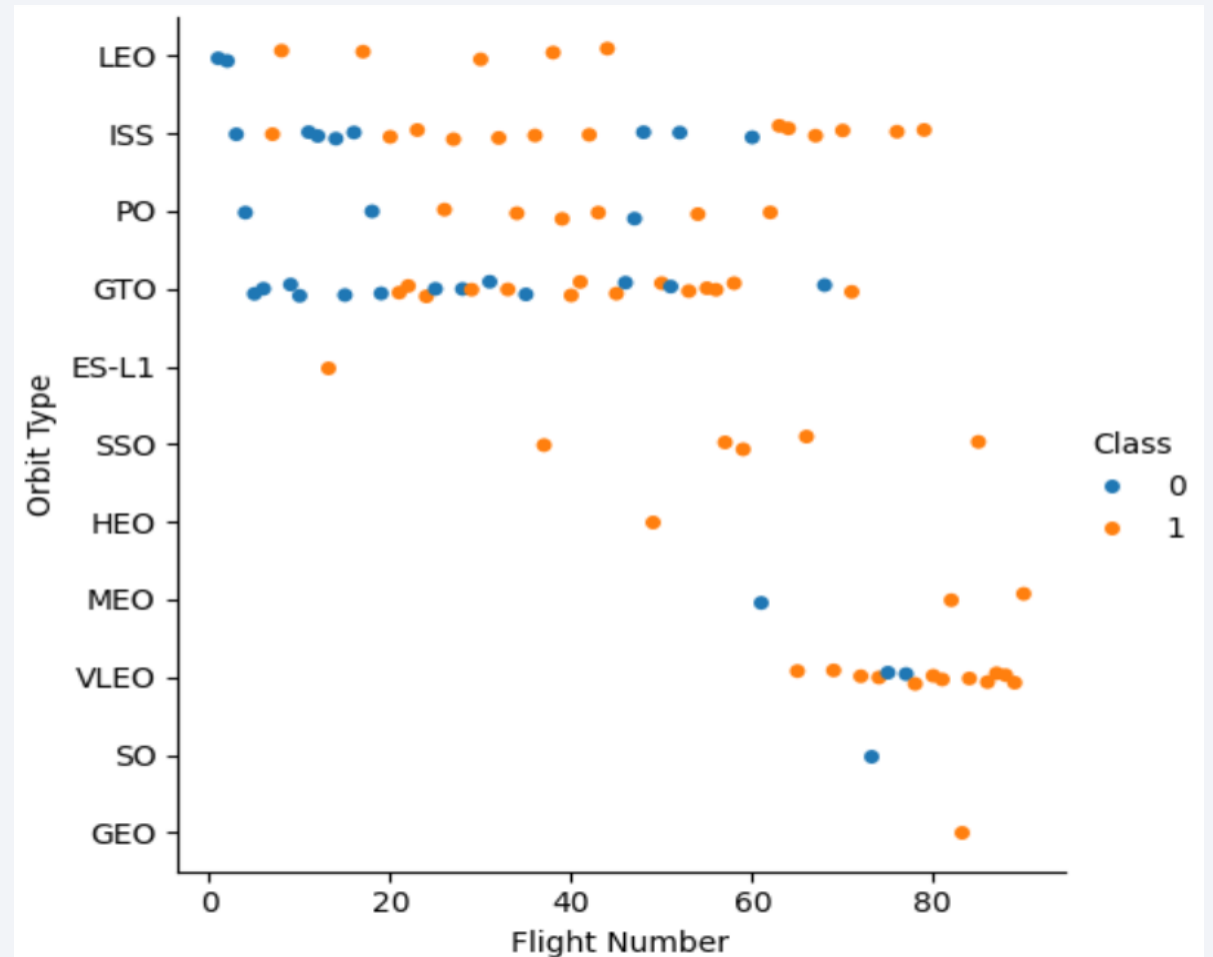
Success Rate vs. Orbit Type

- This plot shows a bar chart for the success rate of each orbit type.
- It indicates that the orbits with the highest success rate are;
 - ES-L1. (Earth-Sun first lagrangian point)
 - SSO. (Sun-Synchronous orbit)
 - HEO. (High Earth Orbit)
 - GEO. (Geostationary Orbit)
- The SO (Heliocentric Orbit) is seen to have the lowest success rate.



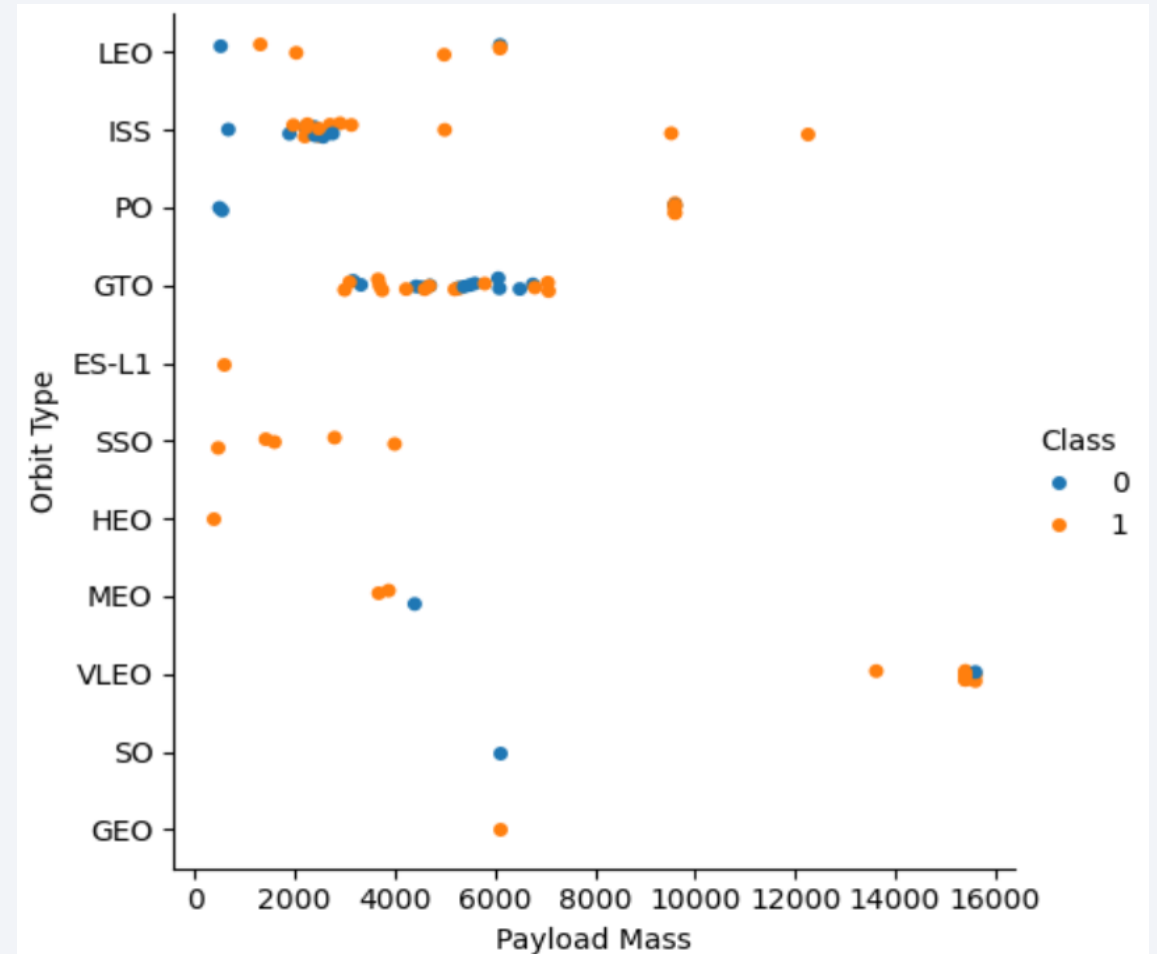
Flight Number vs. Orbit Type

- This figure shows a scatter plot of Flight number vs. Orbit type.
- It reveals that LEO shows an increase in success rate with increase in flight number.
- It also shows that in GTO, there appears to be no significant relationship between flight number and success rate.
- It also indicates that the high success rate in orbits GEO, HEO & ES-L1 is attributed to there being only one flight. SS0



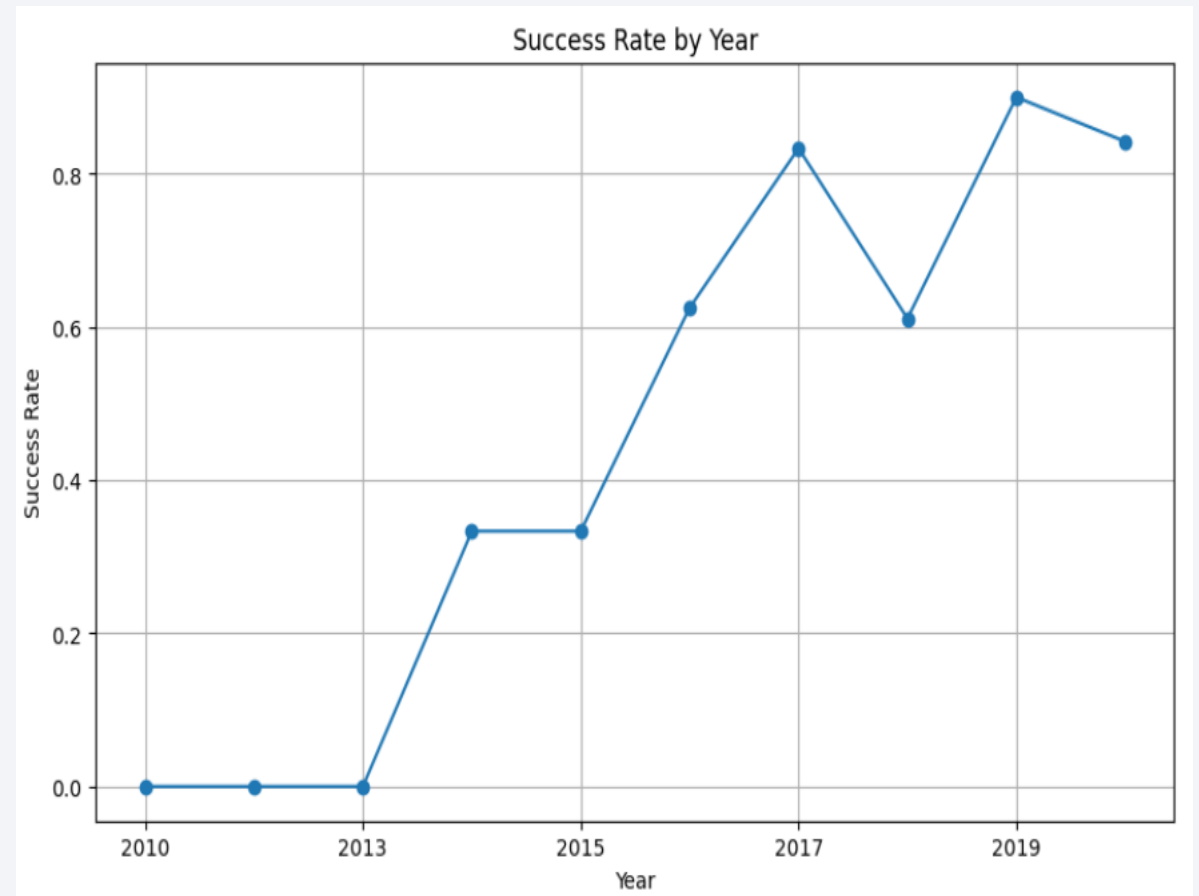
Payload vs. Orbit Type

- This plot shows a scatter chart of payload vs. orbit type which reveals that;
- GTO shows no clear relationship between payload mass and success rate.
- Orbit types PO, ISS & LEO have a stronger success rate with higher payloads.
- VLEO appears to only be associated with significantly higher payloads.



Launch Success Yearly Trend

- This figure shows a line chart of yearly average success rate which reveals that;
- landings prior to 2013 were all unsuccessful.
- There has been consistent increase in success rate over the years apart from a slight decrease in 2018.
- Landings conducted after 2016 all have more than a 50% chance of being successful.



All Launch Site Names

- In this case, we apply the UNIQUE function to return all the unique values in the launch site column.

```
%sql SELECT DISTINCT Launch_site FROM SPACEXTBL
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```


Launch Site Names Begin with 'CCA'

- For this instance, we use the LIKE function with 'CCA%' to filter string values beginning with CCA, as well as the LIMIT function to retrieve only five records.

```
%sql SELECT LAUNCH_SITE FROM SPACEXTBL WHERE LAUNCH_SITE LIKE 'CCA%' LIMIT 5;
```

Launch_Site
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40

Total Payload Mass

- In this scenario, we apply the SUM function to the payload mass column which returns the total.

```
%%sql
SELECT SUM(PAYLOAD_MASS_KG_)
FROM SPACEXTBL
WHERE CUSTOMER = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
Done.
```

SUM(PAYLOAD_MASS_KG_)
45596

Average Payload Mass by F9 v1.1

- In this case, we apply the AVG function to find the average in the payload mass column, followed by the Where function to filter the booster versions and only retrieve F9 v1.1.

```
%%sql
SELECT AVG(PAYLOAD_MASS_KG_)
FROM SPACEXTBL
WHERE BOOSTER_VERSION = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
Done.
```

<u>AVG(PAYLOAD_MASS_KG_)</u>
2928.4

First Successful Ground Landing Date

- Here, we apply the MIN function to the date column to find the earliest date followed a WHERE function which filters only successful ground pad landing records.

```
%%sql
SELECT MIN(DATE) AS "First successful ground pad landing"
FROM SPACEXTBL
WHERE LANDING_OUTCOME = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
First successful ground pad landing
```

```
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- In this scenario, we use the WHERE function to filter successful drone ship landing records followed by the AND & BETWEEN function to filter records with payload mass ranging from 4000Kg to 6000Kg.

```
%%sql
SELECT BOOSTER_VERSION
FROM SPACEXTBL
WHERE LANDING_OUTCOME = 'Success (drone ship)'
  AND PAYLOAD_MASS_KG_ BETWEEN 4000 AND 6000;
```

```
* sqlite:///my_data1.db
```

Done.

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- In this case, we apply the COUNT function to find the total number of missions followed by the GROUP BY statement to separate the records depending on the mission outcome.

```
%%sql
SELECT MISSION_OUTCOME, COUNT(MISSION_OUTCOME) AS TOTAL_NUMBER
FROM SPACEXTBL
GROUP BY MISSION_OUTCOME;
```

```
* sqlite:///my_data1.db
Done.
```

Mission_Outcome	TOTAL_NUMBER
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- Here, we start by applying the DISTINCT statement to retrieve unique booster versions followed by a WHERE clause where we add a subquery that selects the maximum payload.

```
%%sql
SELECT DISTINCT BOOSTER_VERSION
FROM SPACEXTBL
WHERE PAYLOAD_MASS_KG_ = (
    SELECT MAX(PAYLOAD_MASS_KG_)
    FROM SPACEXTBL
);
```

```
* sqlite:///my_data1.db
Done.
Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7
```


2015 Launch Records

- In this scenario, we apply the WHERE function to filter the records for failed drone ship landings and limit the findings to the year 2015.

```
%%sql
SELECT substr(Date, 6, 2) AS Month,
       Landing_Outcome,
       Booster_Version,
       Launch_Site
FROM SPACEXTBL
WHERE substr(Date, 0, 5) = '2015'
      AND Landing_Outcome = 'Failure (drone ship)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Here, we make use of the WHERE and BETWEEN functions to filter the specified dates, followed by the GROUP BY function which separates the records depending on the landing outcome.
- Finally, we use the ORDER BY function to have our findings presented in descending order.

```
%%sql
SELECT Landing_Outcome,
       COUNT(*) AS Outcome_Count
FROM SPACEXTBL
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC;
```

```
* sqlite:///my_data1.db
Done.
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

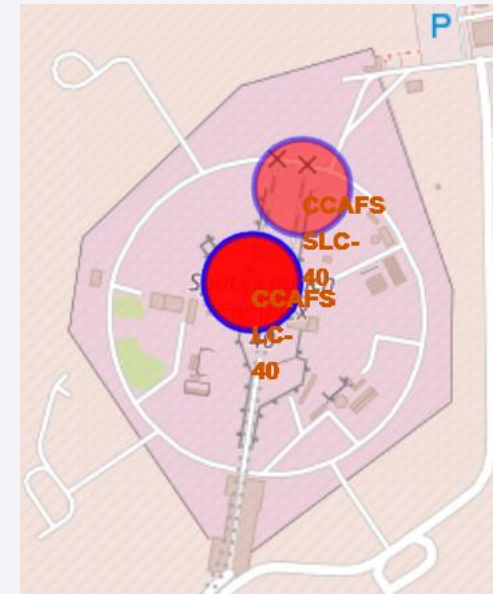
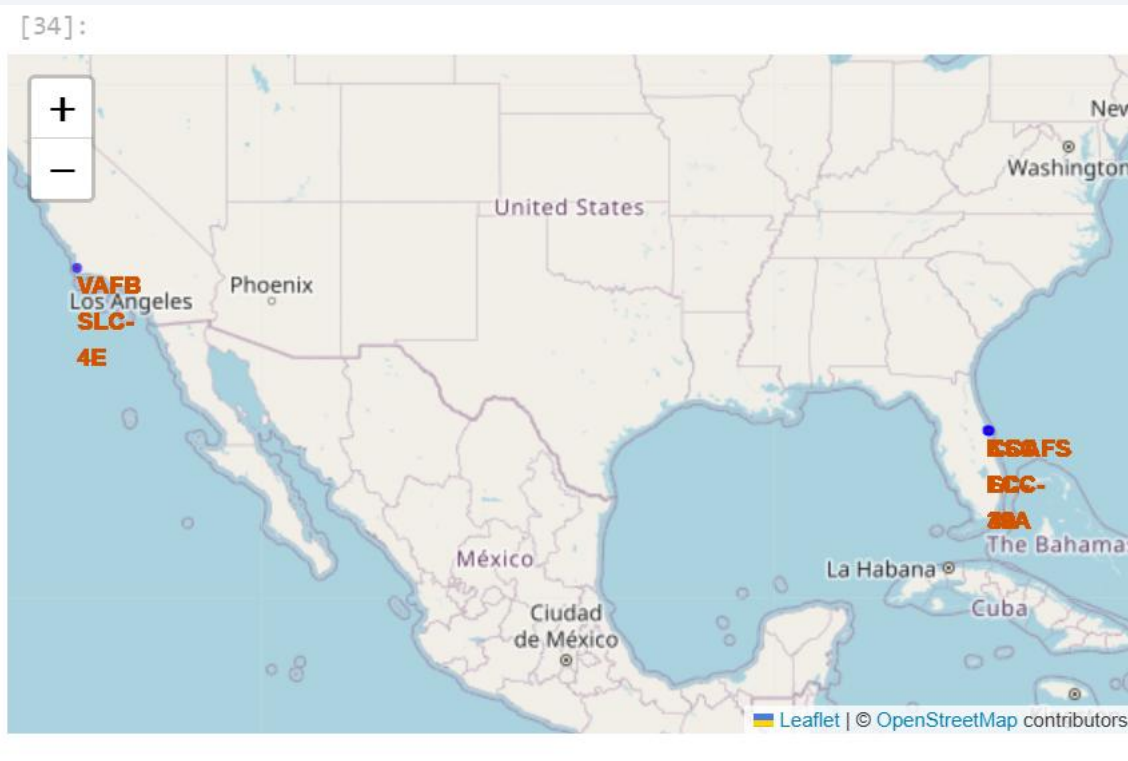
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

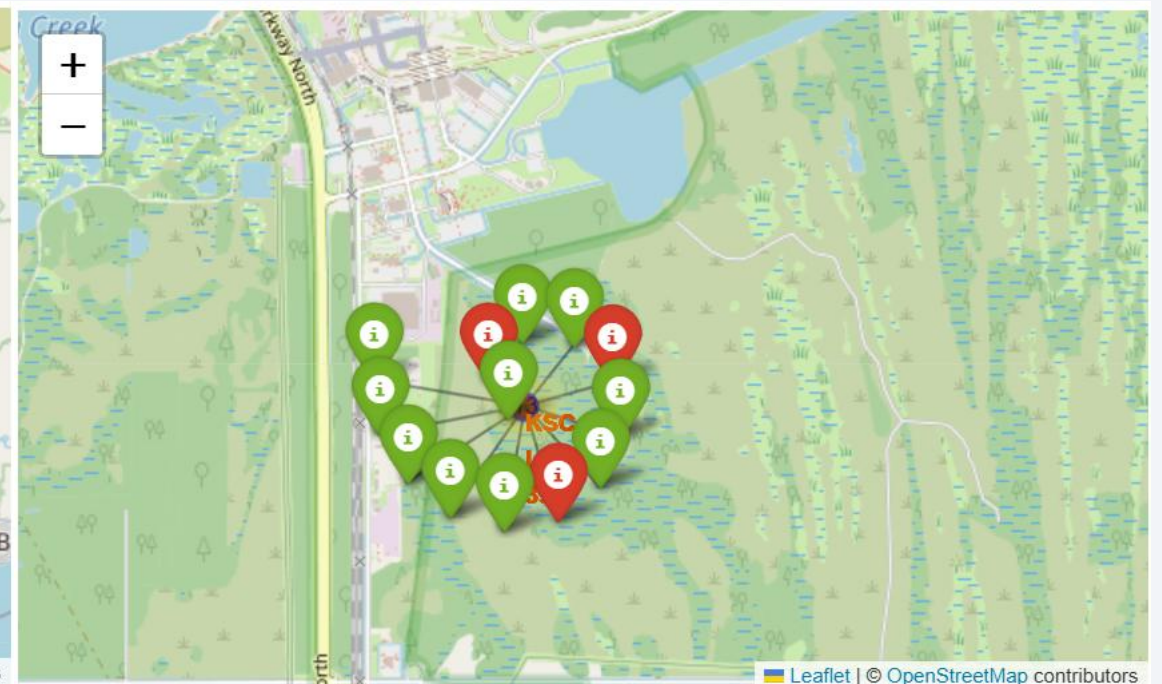
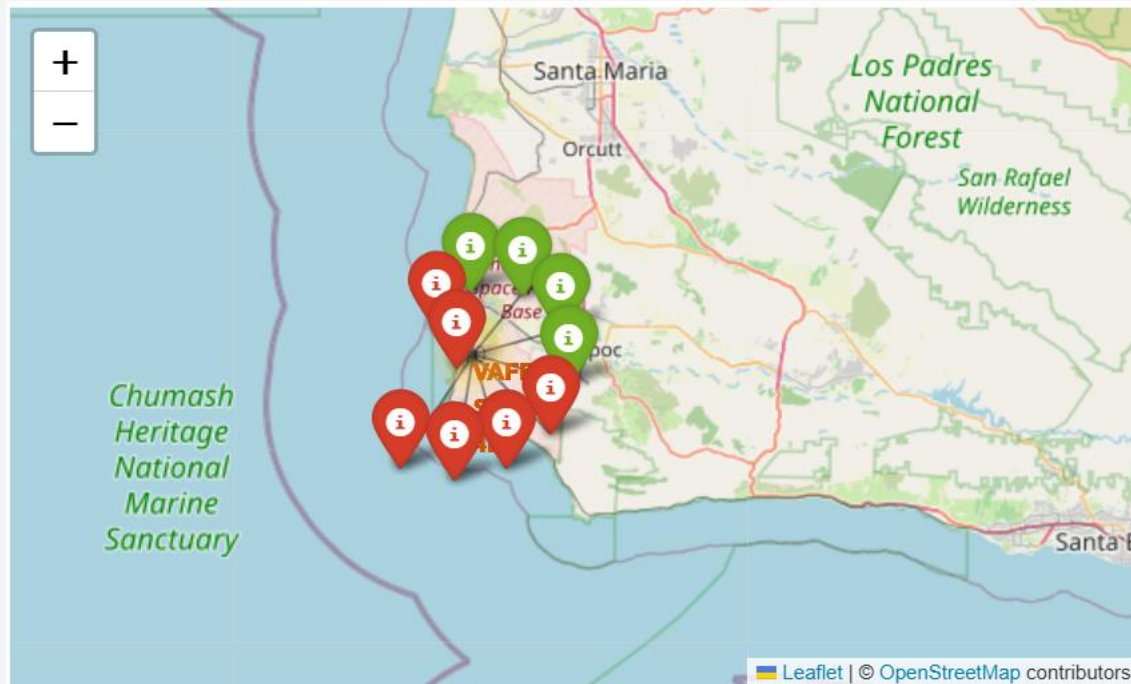
Launch Site Locations

- We can observe that all launch sites are strategically placed along the coastlines of California and Florida.



Launch Outcomes for Each Site

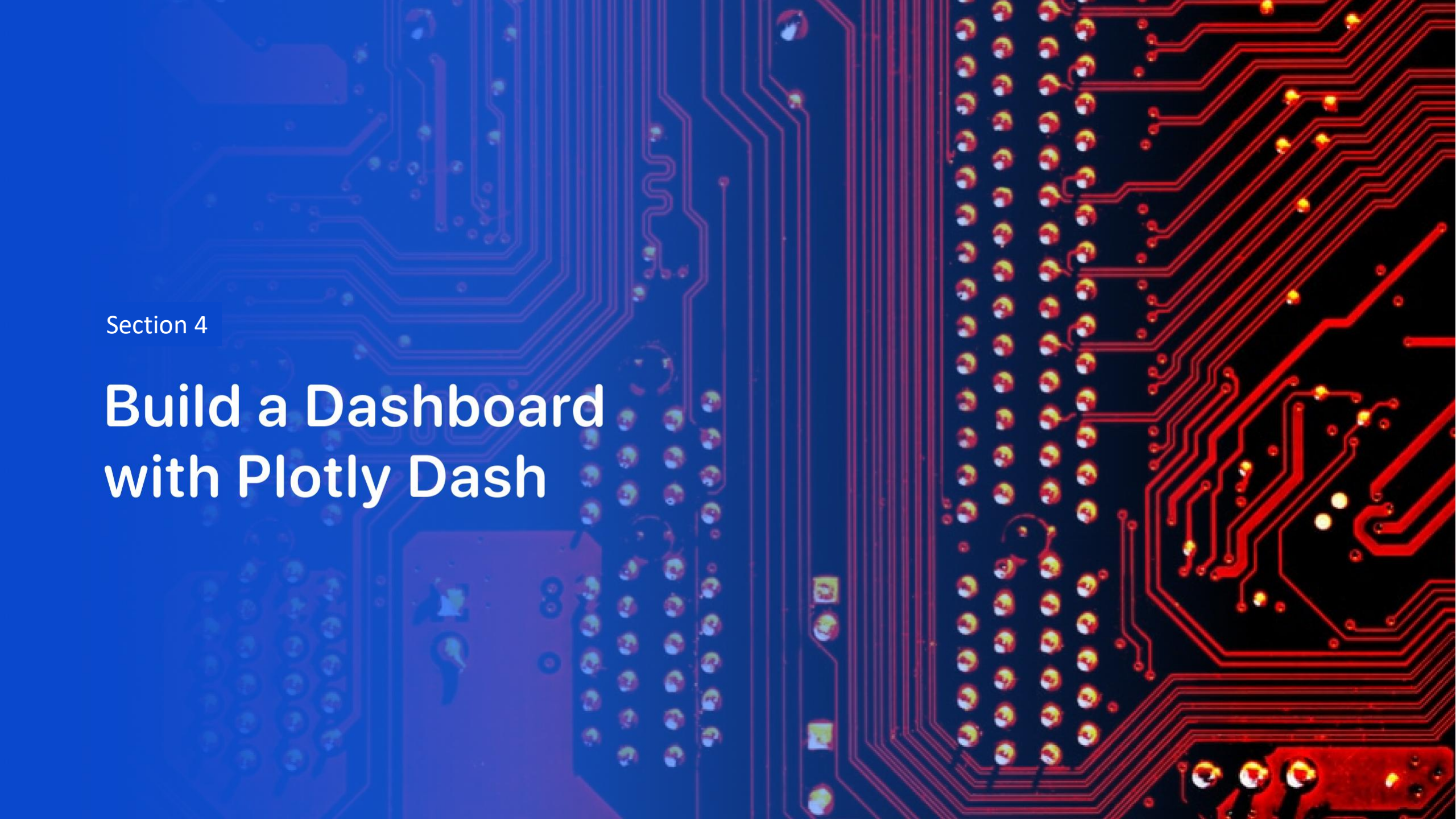
- In order to display this information on the map, we first group the launches into clusters then assign either green or red markers relative to mission outcome. (Green = Success, Red = Failure)



Proximity to Landmarks



- Here, we discover that launch site CCAFS SLC-40 is only 0.86KM from the coastline of Florida.
- On the other hand, Launch site KSC LC-39A is 0.7KM away from the railway line.

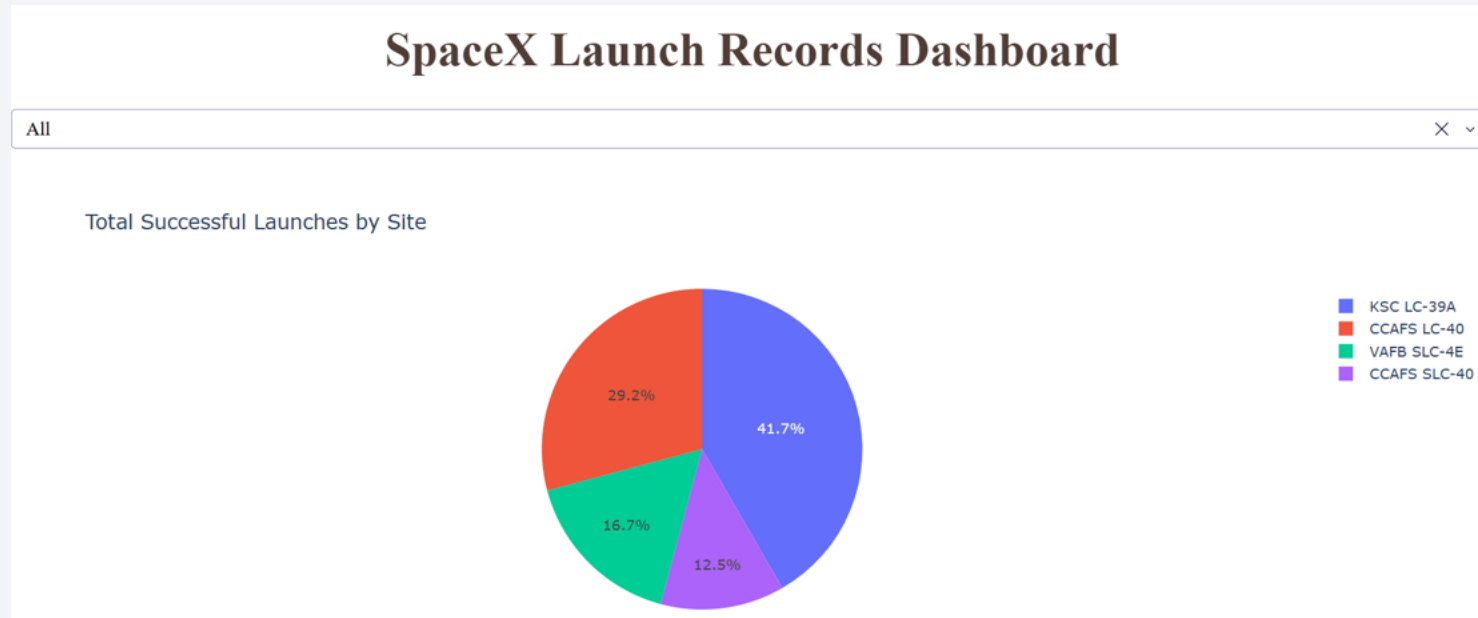


Section 4

Build a Dashboard with Plotly Dash

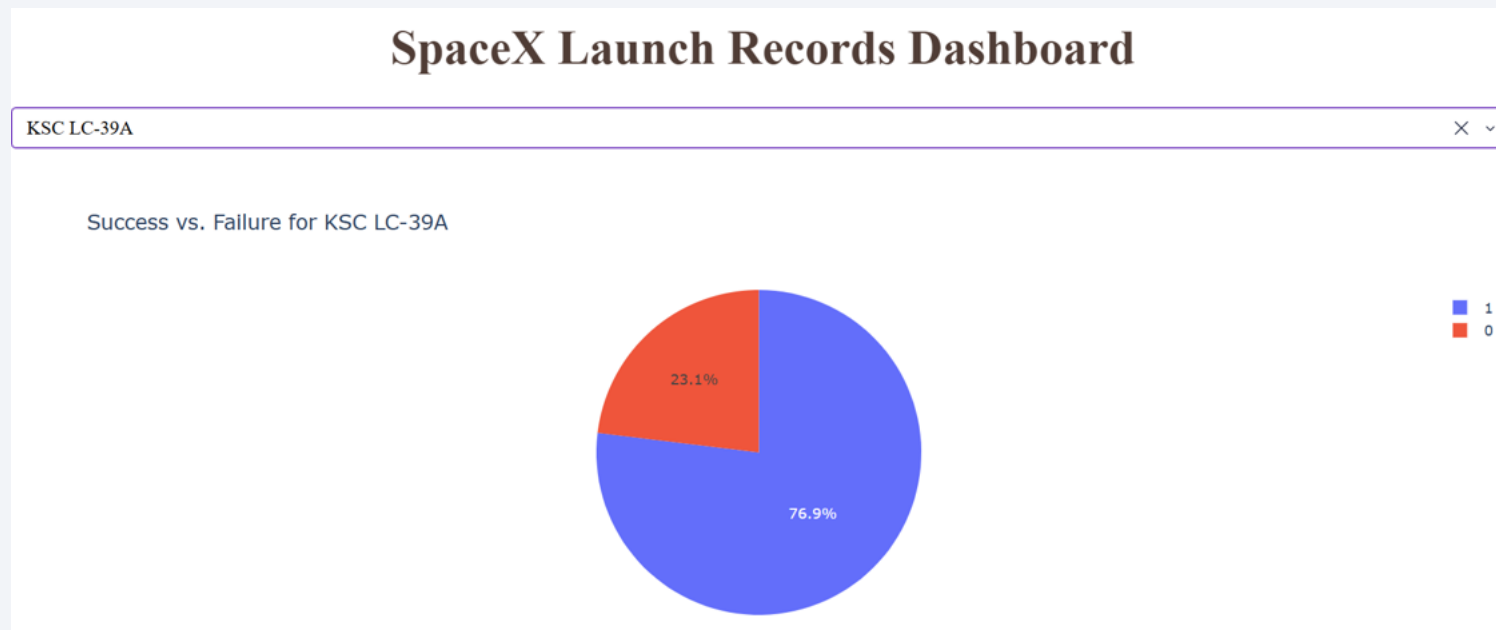
Launch Success Count: All Sites

- We can observe that launch site KSC LC-39A has the highest success rate with 41.7% of the total successful launches.

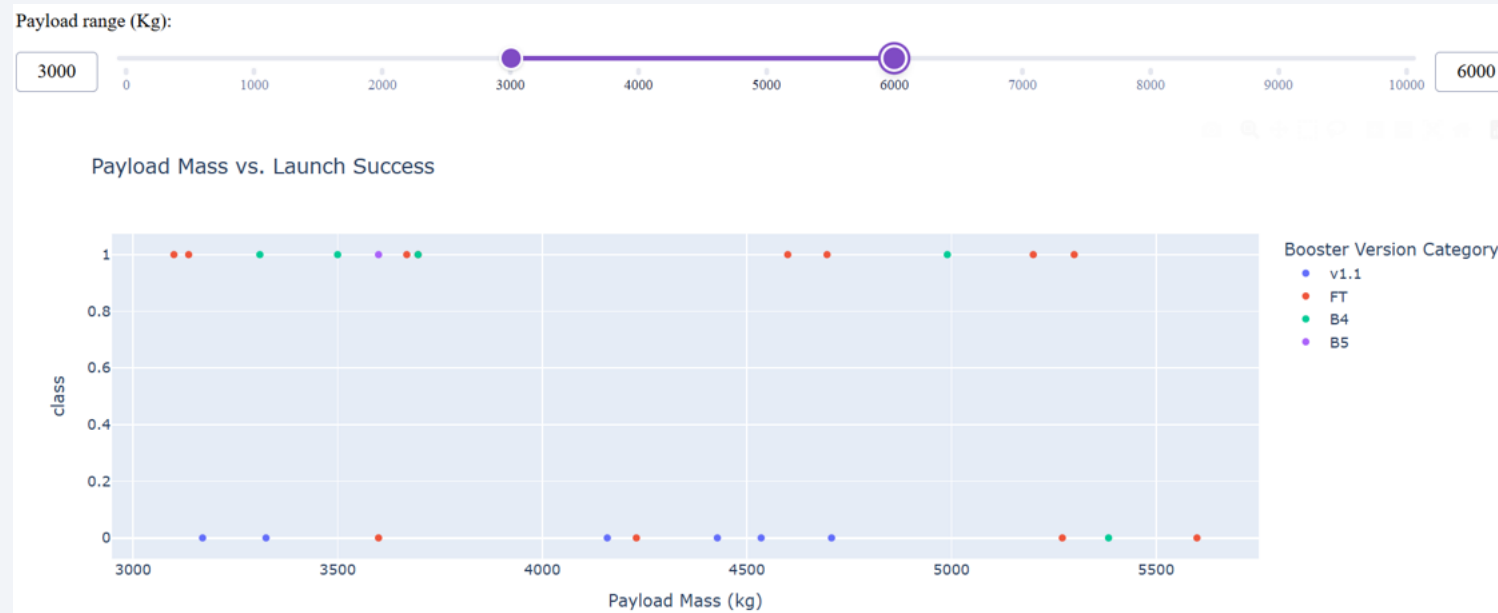


Launch Site with Highest Success Ratio

- Launch site KSC LC-39A has the highest success ratio with 76.9% successful launches and 23.1% failures.



Launch Outcome Given Payload Range



- Given a payload range of 3,000Kg to 6,000Kg, booster version FT shows the most successful mission outcomes. It is followed by versions B4 and B5 whereas version V1.1 is seen to have the most mission failures in this range.

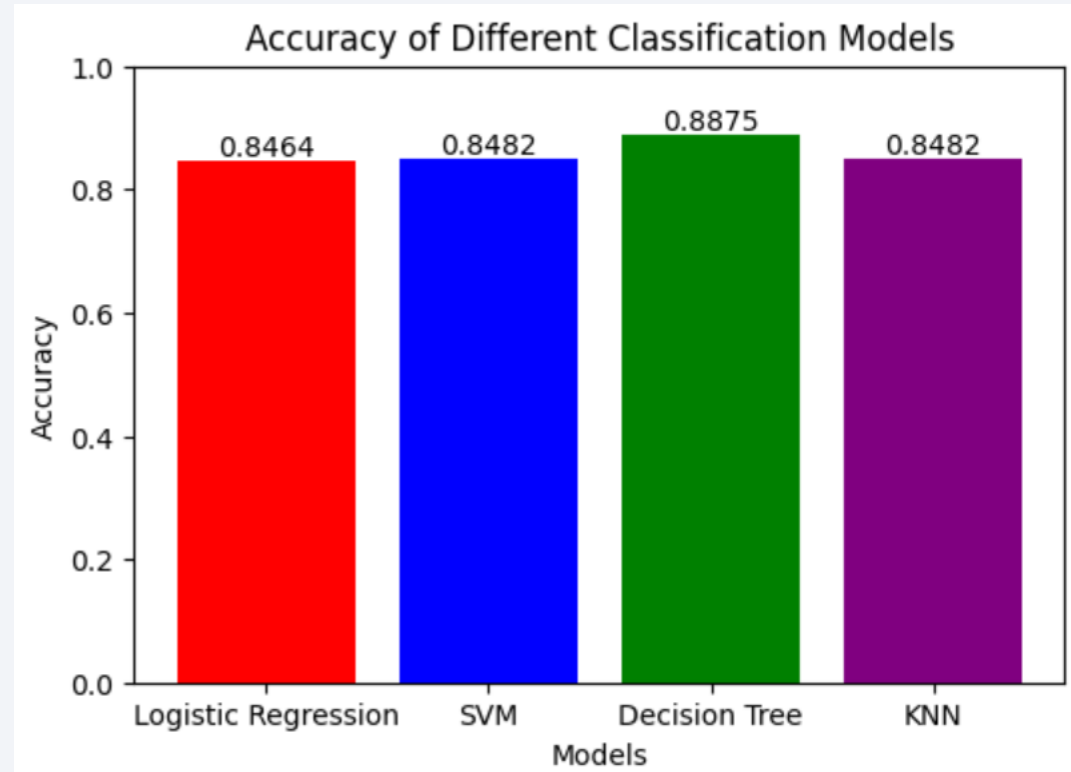


Section 5

Predictive Analysis (Classification)

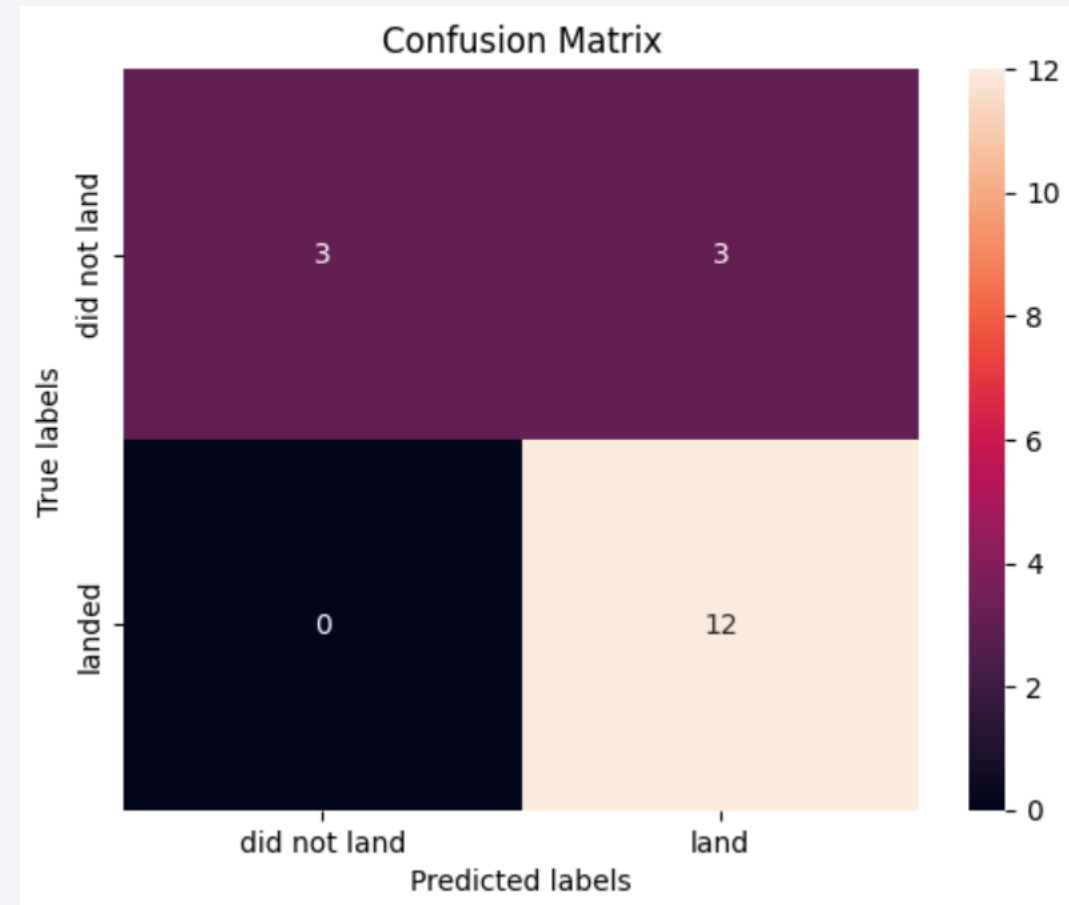
Classification Accuracy

- This bar chart indicates the performance in terms of accuracy of our four models.
- We can infer that despite all models performing quite similarly, Decision Tree Classifier achieves the highest accuracy of 88.75%.



Confusion Matrix

- This confusion matrix is created from the Decision Tree Classifier, which is our best performing model.
- It indicates that only 3 out of 18 instances misclassified as false positives.
- This means that 12 landings actually occurred whereas 3 did not land.



Conclusions

- Despite success rate being low in earlier years (2010-2013), a consistent increase is observed over time. This can be attributed to more experimentation with increase in flight numbers as well as technological advancements in recent years.
- Mission success rate varies depending on the orbit type. GEO, HEO, SSO and ES-L1 have the highest success rate (100%) of all the orbits. On the other hand, PO, LEO, ISS have more successful mission outcomes with heavy loads.
- The different launch sites also played a key role in mission outcome with KSC LC-39A having the most successful launches at 41.7% of all successful launches.
- The selected predictive models all showed good levels of performance with the Decision Tree Classifier emerging on top with the highest accuracy.

Appendix

- Sample of SpaceX dataset used in this project.

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0

Appendix

- Sample of transformed dataset used in model training.

	FlightNumber	PayloadMass	Flights	Block	ReusedCount	Orbit_ES-L1	Orbit_GEO	Orbit_GTO	Orbit_HEO	Orbit_ISS	...	Serial_B1058	Serial_B1059
0	1.0	6104.959412	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0
1	2.0	525.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0
2	3.0	677.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0	...	0.0	0.0
3	4.0	500.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0
4	5.0	3170.000000	1.0	1.0	0.0	0.0	0.0	1.0	0.0	0.0	...	0.0	0.0

Serial_B1060	Serial_B1062	GridFins_False	GridFins_True	Reused_False	Reused_True	Legs_False	Legs_True
0.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0
0.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0
0.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0
0.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0
0.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0

Thank you!

